



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO K.M. FUEELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

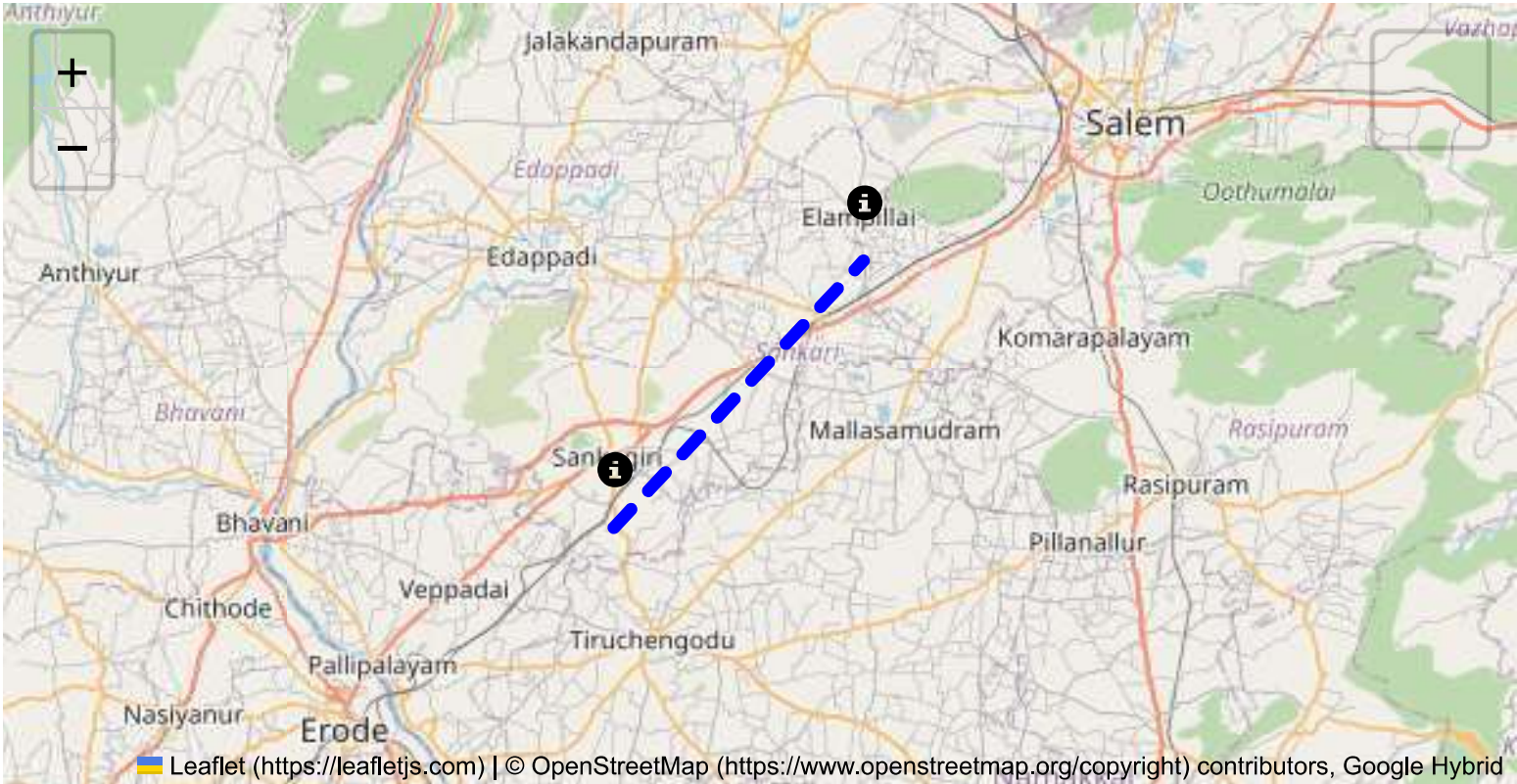
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

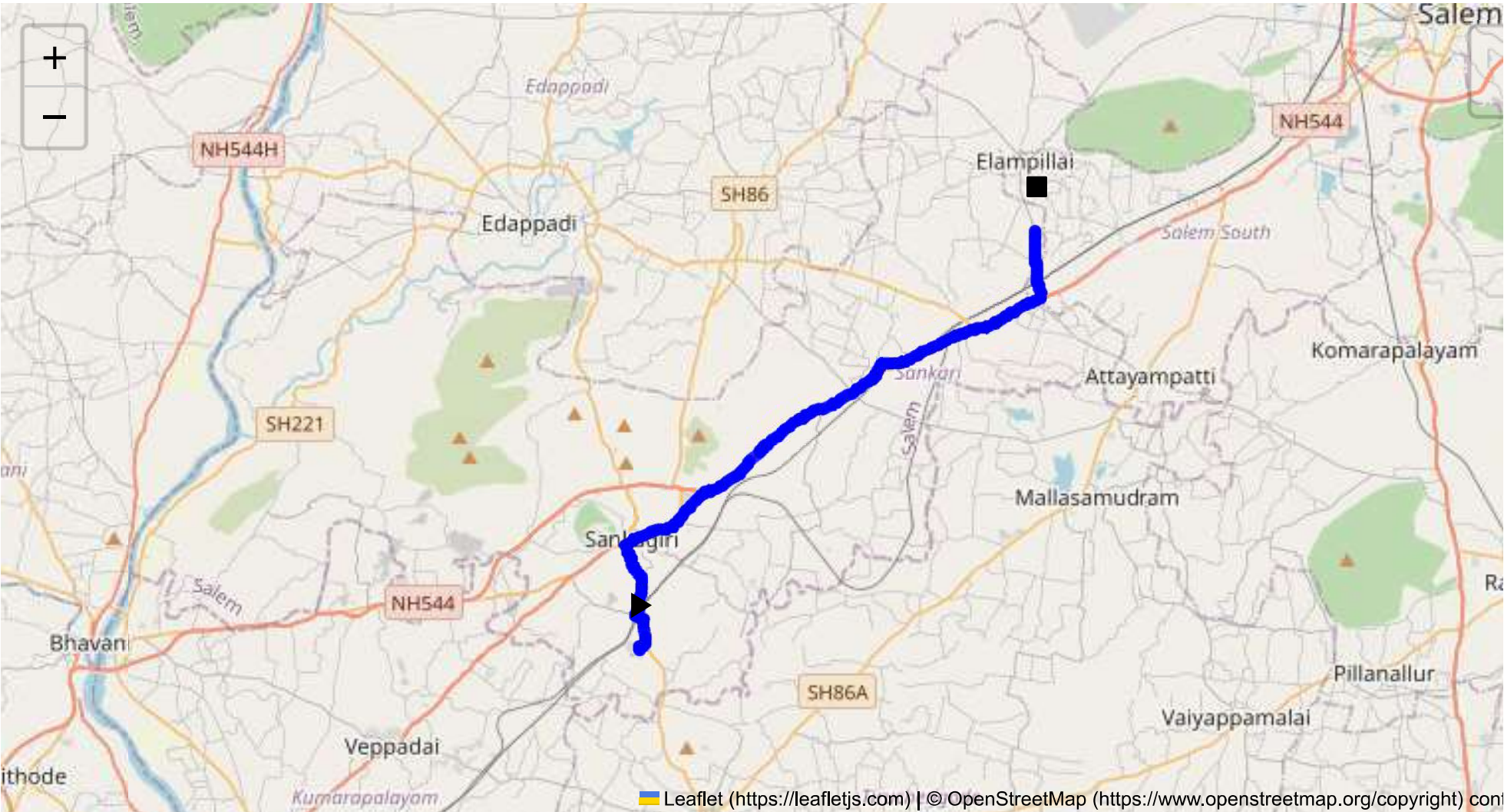
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 25.88 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.580876, 78.010185)



Welcome to the Journey Risk Management Study

To provide a comprehensive safety analysis of the route from Sangagiri to Vembadithalam via the specified locations, several factors need to be considered. Since the route spans approximately 25.88 kilometers, here's a detailed examination based on your points of interest:

1. Route Overview:

- This route predominantly utilizes regional roads connecting Sangagiri to Vembadithalam. It likely traverses rural areas with a mix of agricultural and semi-urban landscapes. Traffic is typically lighter than on highways, but road quality might vary.

2. Weather Conditions and Hazards:

- Tamil Nadu experiences a tropical climate with hot summers, a monsoon season from June to November, and drier winters. Potential weather-related hazards include heavy rainfall during the monsoon, leading to possible road flooding or reduced visibility. Additionally, extreme heat in summer could affect road conditions and vehicle performance.

3. Traffic Patterns:

- Traffic is generally moderate, but peak hours could coincide with morning and evening rush hours. Market days or local festivals can also cause unexpected congestion. The route may have less traffic overall compared to major urban roads but can experience slow-moving agricultural vehicles.

4. Road Quality and Infrastructure:

- Road quality can vary from well-paved sections to less maintained stretches with potholes and uneven surfaces. Watch for narrow bridges or poorly marked intersections. Signage may be limited, requiring drivers to rely more on navigation systems.

5. Alternative Routes:

- In case of emergencies, detours could involve neighboring regional roads or connecting back to main highways such as the NH-44, providing better infrastructure but potentially involving longer travel distances.

6. Local Regulations:

- Tamil Nadu has specific regulations for the transport of hazardous materials, including restrictions on transporting such materials through populated areas and requirements for specific permits. Compliance with these regulations is mandatory.

7. Historical Incidents:

- Historical data on heavy vehicle incidents or hazardous material spills in this region may not be well-documented publicly, but drivers should be cautious around accident-prone intersections or curves.

8. Environmental Considerations:

- The route may pass through areas with agricultural significance, necessitating precautions to prevent contamination. If passing near water bodies, strict adherence to spill prevention measures is crucial.

9. Communication Coverage:

- While mobile network coverage in Tamil Nadu is generally good, rural stretches may experience occasional signal dropouts. It's advisable to have offline navigation tools and planned communication check-in points.

10. Emergency Response Times:

- Response times could vary significantly, with faster times near urban areas and delays in remote stretches. Local hospitals should be noted as waypoints, and emergency contacts saved for quick access.

11. Overall Summary of Risk Assessment:

- This route involves moderate risk primarily due to varying road conditions, potential weather impacts, and traffic patterns. Attention to weather forecasts, road advisories, adherence to safety regulations, and preparedness for communication outages are vital to mitigate risk.

These points collectively outline potential hazards and considerations for safely navigating this route. It's important for truck drivers to continuously update their knowledge and awareness of local conditions and regulations before embarking on the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	High	11.48369, 77.88781	15 KM/Hr
7	Turn	Medium	11.48370, 77.88787	30 KM/Hr
8	Turn	Medium	11.49149, 77.89601	30 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.55622, 78.00774	30 KM/Hr
11	Turn	High	11.55839, 78.01288	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47407, 77.86839



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48369, 77.88781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48370, 77.88787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55622, 78.00774

